COMPACT SUPPORT DUALITY IS COMPATIBLE WITH RESTRICTING TO AN OPEN SUBSCHEME

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0G5B (sources/stacks-duality/pinned-original-source.tex line 9190).

Allowed standard prerequisites (editorial).

- Compact support duality for a separated finite type scheme over a field. Same chapter, Lemma labelled lemma-duality-compact-support, cited at proof lines 9209-9213 and throughout. Supplies the isomorphism whose compatibility is asserted.
- The covariance maps for the lower shriek along an open subscheme and the choice of the smaller ideal. Same chapter, Remark labelled remark-covariance-open-lower-shriek and Remark labelled remark-covariance-open-j-lower-shriek, cited at proof lines 9199-9203 and 9222-9226. Fix the two vertical arrows and the compatible system of ideals.
- Functoriality of the internal Hom evaluation isomorphism. Cohomology, Lemma labelled cohomology-lemma-internal-hom-evaluate, cited at proof lines 9256-9260. Gives the commutativity of the first intermediate square.
- Duality for a proper morphism over a field. Same chapter, Lemma labelled lemma-duality-proper-over-field, cited at proof lines 9236-9240. Provides the duality isomorphism whose functoriality closes the proof.

Source reference. Original lemma 0G5A: sources/stacks-duality/pinned-original-source.tex, source lines 9116–9116. Same-chapter pointer into the author environment lemma at source line 9116; invoked at proof line 9211 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original reference 0G56: sources/stacks-duality/pinned-original-source.tex, source lines 8923–8923. Same-chapter pointer into the author environment remark at source line 8923; invoked at proof line None of the counted proof.

Date: stacks-project:duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original reference 0G4Y: sources/stacks-duality/pinned-original-source.tex, source lines 8613–8613. Same-chapter pointer into the author environment remark at source line 8613; invoked at proof line 9223 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A8U: sources/stacks-chow/tags.txt, source lines 12927–12927. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 9254 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *With notation as in Lemma 0G5A suppose $U' \subset U$ is an open subscheme. Then the diagram*

$$\begin{array}{ccc} \mathrm{Hom}_k(H^i(U, K), k) & \longrightarrow & H_c^{-i}(U, R\mathcal{H}om_{\mathcal{O}_U}(K, \omega_{U/k}^\bullet)) \\ \uparrow & & \uparrow \\ \mathrm{Hom}_k(H^i(U', K|_{U'}), k) & \longrightarrow & H_c^{-i}(U', R\mathcal{H}om_{\mathcal{O}_{U'}}(K, \omega_{U'/k}^\bullet)) \end{array}$$

is commutative. Here the horizontal arrows are the isomorphisms of Lemma 0G5A, the vertical arrow on the left is the contragredient to the restriction map $H^i(U, K) \to H^i(U', K|_{U'})$, and the right vertical arrow is Remark 0G56 (see discussion before the lemma).

Proof. We strongly urge the reader to skip this proof. Choose X and M as in the proof of Lemma 0G5A. We are going to drop the subscript $\mathcal{O}_X$ from $R\mathcal{H}om$ and $\otimes^{\mathbf{L}}$. We write

$$H^i(U, K) = \mathrm{colim}\, H^i(X, R\mathcal{H}om(\mathcal{I}^n, M))$$

and

$$H^i(U', K|_{U'}) = \mathrm{colim}\, H^i(X, R\mathcal{H}om((\mathcal{I}')^n, M))$$

as in the proof of Lemma 0G5A where we choose $\mathcal{I}' \subset \mathcal{I}$ as in the discussion in Remark 0G4Y so that the map $H^i(U, K) \to H^i(U', K|_{U'})$ is induced by the maps $(\mathcal{I}')^n \to \mathcal{I}^n$. We similarly write

$$H_c^i(U, R\mathcal{H}om(K, \omega_{U/k}^\bullet)) = \lim H^i(X, R\mathcal{H}om(M, \omega_{X/k}^\bullet) \otimes^{\mathbf{L}} \mathcal{I}^n)$$

and

$$H_c^i(U', R\mathcal{H}om(K|_{U'}, \omega_{U'/k}^\bullet)) = \lim H^i(X, R\mathcal{H}om(M, \omega_{X/k}^\bullet) \otimes^{\mathbf{L}} (\mathcal{I}')^n)$$

so that the arrow $H_c^i(U', R\mathcal{H}om(K|_{U'}, \omega_{U'/k}^\bullet)) \to H_c^i(U, R\mathcal{H}om(K, \omega_{U/k}^\bullet))$ is similarly deduced from the maps $(\mathcal{I}')^n \to \mathcal{I}^n$. The diagrams

$$\begin{array}{ccc} R\mathcal{H}om(M, \omega_{X/k}^\bullet) \otimes^{\mathbf{L}} \mathcal{I}^n & \longrightarrow & R\mathcal{H}om(R\mathcal{H}om(\mathcal{I}^n, M), \omega_{X/k}^\bullet) \\ \uparrow & & \uparrow \\ R\mathcal{H}om(M, \omega_{X/k}^\bullet) \otimes^{\mathbf{L}} (\mathcal{I}')^n & \longrightarrow & R\mathcal{H}om(R\mathcal{H}om((\mathcal{I}')^n, M), \omega_{X/k}^\bullet) \end{array}$$

commute because the construction of the horizontal arrows in Cohomology, Lemma 0A8U is functorial in all three entries. Hence we finally come down to the assertion

that the diagrams

$$\begin{array}{ccc} \mathrm{Hom}_k(H^i(X, R\mathcal{H}om(\mathcal{I}^n, M)), k) & \longrightarrow & H^{-i}(X, R\mathcal{H}om(R\mathcal{H}om(\mathcal{I}^n, M), \omega^\bullet_{X/k})) \\ \uparrow & & \uparrow \\ \mathrm{Hom}_k(H^i(X, R\mathcal{H}om((\mathcal{I}')^n, M)), k) & \longrightarrow & H^{-i}(X, R\mathcal{H}om(R\mathcal{H}om((\mathcal{I}')^n, M), \omega^\bullet_{X/k})) \end{array}$$

commute. This is true because the duality isomorphism

$$\mathrm{Hom}_k(H^i(X, L), k) = \mathrm{Ext}^{-i}_X(L, \omega^\bullet_{X/k}) = H^{-i}(X, R\mathcal{H}om(L, \omega^\bullet_{X/k}))$$

is functorial for L in $D_{QCoh}(\mathcal{O}_X)$. □